

Liquidity and the Post-Earnings-Announcement-Drift

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Abstract

The post-earnings-announcement-drift is a long standing anomaly that is in conflict with weak-form market efficiency. This paper documents that the post-earnings-announcement-drift occurs mainly in the highly illiquid stocks. A trading strategy that goes long the high earnings surprise stocks and short the low earnings surprise stocks provides a return of 0.24% in the most liquid stocks and 1.79% per month in the most illiquid stocks. The illiquid stocks have high trading costs and market impact costs. Using a multitude of estimates we find that transaction costs account for anywhere from 66% to 100% of the paper profits from the long-short strategy designed to exploit the earnings momentum anomaly.

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1 Introduction

In a seminal paper, Fama (1998) once again makes the case for market efficiency, arguing that the null should still be one of market efficiency. However, Fama does concede that the existence of two robust and persistent anomalies still pose challenges to the efficient markets paradigm. One of these anomalies is the the post-earnings-announcement-drift or earnings momentum.¹ Earnings momentum refers to the fact that firms reporting unexpectedly high earnings subsequently outperform firms reporting unexpectedly low earnings. The predictable drift in stock prices following earnings announcements is one of the longest-standing anomalies. Schwert (2003) has argued that a number of market anomalies have typically disappeared, reversed or attenuated following their discovery. The post-earnings-announcement-drift, along with price momentum, is still robust after its initial discovery. Ball and Brown (1968) were the first to note that stock returns continue to drift in the direction of earnings surprises for several months after the earnings are announced. Since then, several studies have confirmed the robustness of the post-earnings-announcement-drift or earnings momentum using more recent data and using data from stock markets other than the US, where the phenomenon was first identified.²

The post-earnings-announcement-drift has been robust over almost four decades and is commonly interpreted as evidence that investors underreact to earnings surprises (Bernard and Thomas, 1990; Ball and Bartov, 1996). Most empirical studies of the post-earnings-announcement drift suggest that the empirical evidence is consistent with market inefficiency and investor irrationality (Bernard and Thomas, 1990; Rangan and Sloan, 1998; Brown and Han, 2000; Mendenhall, 2005). However, it is difficult to understand why stock prices appear to not respond completely and immediately to information as visible and freely available as publicly announced earnings. Chordia and Shivakumar (2005) document that a portfolio that is long in stocks with the highest earnings surprises and short in stocks with the lowest earnings surprises provides a return of 90 basis points per month or over 10% annually. Such large profits, from a simple long-short trading strategy, over a period of almost four decades point to a violation of the weak-form market efficiency as defined by Fama (1970).

¹The other anomaly is the price momentum, documented by Jegadeesh and Titman (1993).

²Foster, Olsen and Shevlin (1984) and Bernard and Thomas (1990), among others, have confirmed the robustness of the Ball and Brown (1968) findings using more recent data. Hew, Skerratt, Strong and Walker (1996) and Booth, Kallunki and Martikainen (1996) have extended the post-earnings-announcement drift evidence to non-US data.

In this paper, our goal is to answer the following question. What has reason behind the profits from an earnings momentum strategy being robust over a period of four decades? Or in other words, do the earnings momentum strategy profits really represent violations of weak form efficiency? We also weigh in on the debate between the behavioral and rational asset pricing models. The recent interest in behavioral models is driven by data that conflicts with the standard frictionless asset pricing models. We introduce frictions in the form of trading costs to evaluate the profitability of trading strategies that exploit the post-earnings-announcement-drift. More specifically, we examine the impact of illiquidity on the trading profits. While liquidity is an elusive concept, most market participants agree that liquidity generally reflects the ability to buy or sell sufficient quantities, quickly, at low trading cost and without impacting the market price too much. Following Amihud (2002), we measure monthly illiquidity as the average of the daily price impacts of the order flow, i.e., the daily absolute price change per dollar of daily trading volume.³ We examine the profitability of the long-short strategy after sorting stocks into decile portfolios based on the Amihud (2002) illiquidity measure.

We find that the post-earnings-announcement-drift is prevalent mainly in stocks that are relatively illiquid. The returns from the long-short strategy increases monotonically from 0.24% for the most liquid stocks to 1.79% per month for the most illiquid. Moreover, we demonstrate that the long-short trading strategy that attempts to profit from the post-earnings-announcement-drift generates high transactions costs and substantial price impact. Specifically, we use the transaction cost estimates of Keim and Madhavan (1997) within a dynamic framework to estimate net returns based on a strategy that buys the high earnings surprise stocks and sells the low earnings surprise stocks. We repeat the analysis of Korajczyk and Sadka (2004) and Chen, Stanzl and Watanabe (2004) to examine market impact costs for a \$5 million portfolio. The evidence conclusively shows that transaction costs consume anywhere from 66% to 100% of the potential profits. Thus, a mutual fund, as small as \$5 million, designed to exploit the post-earnings-announcement-drift loses anywhere from 66% to 100% of the potential profits to transaction costs. While the presence of statistically significant earnings momentum in individual security returns is undeniable, it is not possible to profit from this predictability. This lack of profitability is consistent with the Jensen's (1978) definition of market efficiency and Rubinstein's (2001) definition of minimally rational markets.⁴

³The Amihud measure has been inspired by the Kyle (1985) lambda, which is a measure of the market impact.

⁴The concept of market efficiency with respect to an information set has been defined by Jensen (1978) as the inability to profit from that information. Rubinstein (2001) defines this as minimally rational markets.

The lack of profitability from a long-short strategy that exploits earnings momentum suggests that the violations of the efficient market hypothesis due to the post-earnings-announcement-drift are not so egregious after all.

This paper is related to a recent strand of the literature that re-examines market anomalies in the context of transactions costs. For instance, Korajczyk and Sadka (2004) have examined the impact of transactions costs on price momentum strategies. Avramov, Chordia and Goyal (2006) have documented that the short-run reversal strategies at the weekly and monthly frequencies are not profitable once transactions costs are taken into account. Hanna and Ready (2006) find that the Haugen and Baker (1996) trading strategies do not provide attractive returns after accounting for transactions costs. The above papers are related to ours in the sense that transactions costs account for a large fraction of the potential profitability of the market anomalies.

The remainder of the paper proceeds as follows. Section 2 presents the data. Section 3 presents the results. A detailed discussion of transactions costs is presented in section 4. Section 5 examines liquidity around earnings announcement dates and section 6 concludes.

2 Data

Our sample consists of all NYSE-AMEX firms with data available on the monthly CRSP and quarterly COMPUSTAT files for the period January 1972 through December 2004. We focus only on common stocks, and eliminate ADRs, REITs, Americus Trust Components, units, and closed-end funds from the sample. In order to avoid the extremely illiquid securities, stocks with prices less than five dollars, have been eliminated from the sample.⁵ Our tests use the standardized unexpected earnings (SUE) to capture the post-earnings-announcement drift (PEAD). In each month, SUE for a firm is computed as the most recently announced quarterly earnings less the earnings four quarters ago. This earnings change is standardized by its standard deviation estimated over the prior eight quarters. The average number of stocks in the sample is 1841.

The Amihud (2002) illiquidity measure is the average daily price impact of order flow

⁵In the context of long-term contrarian investment strategies, Ball, Kothari, and Shanken (1995) show that microstructure issues can create severe biases amongst low-priced stocks. We obtain qualitatively similar results when stocks with prices less than one dollar are eliminated from the sample.

and is computed as the absolute price change per dollar of daily trading volume:

$$ILLIQ_{it} = \frac{1}{D_{it}} \sum_{t=1}^{D_{it}} \frac{|R_{itd}|}{DVOL_{itd}} * 10^6, \quad (1)$$

where R_{itd} is the daily return, $DVOL_{itd}$ is the dollar trading volume of stock i on day d in month t , and D_{it} is the number of days in month t for which data is available for stock i . We compute the Amihud illiquidity measure at the monthly frequency. We require at least ten days with trades each month. Since turnover has often been used as a measure of liquidity⁶ one may expect the illiquidity measure to be related to turnover. However, the cross-sectional correlation between turnover and illiquidity is only -0.08 . The information in the Amihud illiquidity measure is not subsumed by that in turnover.⁷

Table 1 presents summary statistics for SUE sorted decile portfolios. In each month, sample firms are sorted into deciles based on the most recent SUE. In each month, and within each portfolio, we average the SUEs of firms constituting the portfolio to obtain the portfolio's SUE. By design, the average SUE increases monotonically from -3.7 for the lowest SUE decile portfolio to 3.2 for the highest SUE decile portfolio. The higher SUE portfolios (six through ten) are more liquid than the lower SUE portfolios (one through five). The Amihud illiquidity measure is 0.64 for the lowest SUE portfolio and 0.43 for the highest SUE portfolio.

The average monthly returns (computed over a six month period after the formation of the decile portfolios) vary almost monotonically from 0.88% per month for the lowest SUE portfolio to 1.62% for the highest SUE portfolio. The difference in the monthly returns between the two extreme decile portfolios is about 0.73% per month. Thus, the post-earnings-announcement-drift strategy of going long the high SUE stocks and short the low SUE stocks results in a return of about 4.4% over the following six months. The monthly risk-adjusted return is a statistically significant 0.74% when the CAPM is used as a pricing model and 0.87% when the Fama-French (1993) model is used. When the Fama-French factors are augmented by a momentum based factor, UMD, the alpha is 0.65% per month. In other words, the raw return of 0.73% per month is marginally lower at 0.65% per month upon adjusting for risk and upon controlling for the impact of price momentum. The standard asset pricing models do not capture the profits from

⁶See, e.g., Brennan, Chordia, and Subrahmanyam (1998) and Chordia, Subrahmanyam, and Anshuman (2001).

⁷Hasbrouck (2003) compares effective and price impact measures estimated from daily data to those from high frequency data and finds that the Amihud (2002) measure is the most highly correlated with trade-based measures.

an earnings momentum strategy.

3 Results

Table 2 presents the average monthly returns on portfolios that are sorted independently on SUE and illiquidity. The returns are, once again, computed over a six month period after the formation of the SUE-illiquidity portfolios. Within each portfolio the returns are equally weighted. There are a total of a hundred portfolios with the 10x10 sort on SUE and illiquidity. In general, the returns increase with SUE and with illiquidity. Consider, for instance the SUE portfolio 10. The returns increase monotonically from 1.23% for the most liquid portfolio to 2.04% for the most illiquid portfolio. Similarly, for the illiquidity portfolio 10, the returns increase monotonically from 0.25% for the lowest SUE portfolio to 2.04% for the highest SUE portfolio.

The difference in returns between the high SUE portfolio and the low SUE portfolio also increases with illiquidity. For the most liquid stocks, this difference is only 0.24% but for the most illiquid stocks this difference is more than seven times higher at 1.79%. This suggests that the profits from a strategy that exploits the PEAD, by buying the high SUE stocks and selling the low SUE stocks, are higher for the more illiquid stocks. In fact, for the four portfolios with highest liquidity, the PEAD strategy profits are less than 50 basis points per month. Only the three most illiquid portfolios have PEAD strategy profits greater than 1% per month.

Conditioning on illiquidity has a large impact on the returns to the portfolio that is long the high SUE stocks and short the low SUE stocks. The long-short portfolio raw returns for the most illiquid stocks are 1.79% and the risk adjusted alphas are 1.74% when using the CAPM for risk adjustment, 1.86% when adjusting with the Fama-French (1993) model and 1.74% when the Fama-French model is augmented by UMD, the factor for momentum. Given that conditioning on liquidity is important, it is natural to ask whether the loading on a liquidity factor explains the above long-short portfolio returns. Thus, we also augment the Fama-French factor by UMD and the Pastor-Stambaugh liquidity factor. We use a value weighted factor that is long stocks with high sensitivity to the Pastor-Stambaugh non-traded liquidity factor and short stocks with low sensitivity to the non-traded liquidity factor.⁸ The risk adjusted long-short portfolio alpha is 1.81% when using the Fama-French factors along with UMD and the above liquidity factor.

⁸We thank Lubos Pastor for the traded liquidity factor.

We have checked that the factor loadings in each of the above models are insignificant suggesting that the long-short portfolio does not load on the standard risk factors.

To summarize, even after adjusting for the standard risk factors, the long-short portfolio returns are well in excess of 1% per month. However, these excess returns obtain only in the highly illiquid stocks. The fact that the PEAD strategy profits are confined to the most illiquid stocks suggests that these profits may not be easily realizable because it is precisely the illiquid stocks that have high transactions costs and high market impact costs.

Given that the portfolios in Table 2 are formed by sorting independently on SUE and illiquidity, it is natural to ask whether the various portfolios are populated. We have checked to ensure that all the portfolios are indeed populated and the results are not being driven by outliers, even in the highly illiquid portfolios.

Table 3 presents descriptive statistics for the various two-way sorted portfolios in Table 2. Specifically, Panel A of Table 3 presents the equally weighted average illiquidity, turnover, dollar trading volume and the decile size rank for each of the portfolios. These characteristics are calculated over the six month holding period after the portfolio formation date. The results show clearly that while the firm characteristics do not change much with SUE, there is considerable variation with illiquidity. Holding SUE constant, firm size, turnover and dollar trading volume decrease with illiquidity. For instance in the SUE portfolio 10, average dollar trading volume (turnover) decreases from \$8.5 million per month (0.77%) for the most liquid stocks to \$0.007 million per month (0.23%) for the most illiquid stocks. The average decile size rank also decreases monotonically from 9.6 to 1.5. The decile size rank is calculated as per the NYSE breakpoints.⁹

Panel B of Table 3 presents the bid-ask spread and the quoted depths of the various portfolios over the period 1988 through 2002. These measures were obtained from transactions data and are available for NYSE stocks only.¹⁰ The proportional effective and the proportional quoted spread increase and depth decreases with the Amihud illiquidity measure. Consider the SUE portfolio 10. The proportional effective (quoted) spread increases from 0.21% (0.32%)¹¹ for the most liquid stocks to 1.80% (2.77%) for the most illiquid stocks while the quoted depth decreases from 701 to 182 shares.

⁹NYSE breakpoints were obtained from the data library at Ken French's website. See <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/>.

¹⁰See Chordia, Roll and Subrahmanyam (2001) for details about this data.

¹¹For NYSE stocks, the effective spread is lower than the quoted spread because of within the spread trading.

In sum, Table 3 presents clear evidence that the illiquid stocks, as per the Amihud illiquidity measure, are smaller and have higher bid-ask spreads, lower trading volumes and lower quoted depths. In section 4, we will obtain more precise estimates of institutional trading costs by relying on Keim and Madhavan (1997), Korajczyk and Sadka (2004) and Chen, Stanzl and Watanabe (2004) for estimates of transaction costs and market impact costs.

3.0.1 Robustness Checks

At this stage, we must ensure that our results are robust.

Is liquidity simply a proxy for size? Prior research has shown that the post-earnings-announcement-drift is more prevalent in small firms. The issue here is whether liquidity proxies for firm size. To ascertain the importance of liquidity we form portfolios by sorting conditionally on size and the Amihud (2002) illiquidity measure. Since size and liquidity are correlated, it is important to sort conditionally rather than independently to ensure that all portfolios are populated. Table 4 presents the results. Consider Panel A where we first sort the sample firms into four size quartiles and then sort firms in each of the four size quartiles into quartiles based on illiquidity. Finally, we sort firms in each size and illiquidity group into quartiles based on SUE. Thus we have a total of 64 portfolios. Consider the smallest size quartile. Except for the smallest SUE stocks, returns generally increase with illiquidity. Moreover, the returns of the highest SUE stocks less the lowest SUE stocks increase with illiquidity from 1.11% to 1.58% per month. In the largest size quartile as well, we see that holding SUE constant, returns increase with illiquidity. However, the difference in returns of the high and the low SUE portfolio decreases with illiquidity.

In Panel B of Table 4 we first sort on illiquidity and then on firm size. Within each size portfolio we further sort on SUE. Across the high as well as the low liquidity portfolios, except for the low SUE stocks that are illiquid, returns generally decrease with size. Also, the returns of the highest SUE stocks less the lowest SUE stocks decrease with firm size. More importantly, for our purposes, regardless of firm size, the returns are generally higher for the more illiquid stocks. Moreover, the returns of the highest SUE stocks less the lowest SUE stocks are also higher when illiquidity is higher. The highest return to a portfolio that is long the highest SUE stocks and short the lowest SUE stocks obtains for the small illiquid stocks suggesting that both size and illiquidity are important determinants of the post-earnings-announcement-drift. All these results

suggest that illiquidity is an important determinant of earnings momentum.

Cross-sectional Tests: In these tests, we confirm that illiquidity has an important impact on the post-earnings-announcement-drift. We rely on cross-sectional asset pricing tests with individual stocks rather than portfolios. Our asset pricing tests extend the approach of Brennan, Chordia, and Subrahmanyam (BCS) (1998). BCS test factor models by regressing risk-adjusted returns on firm-level attributes such as size, book-to-market, and turnover. Under the null of exact pricing, such attributes should be statistically insignificant in the cross-section. The BCS methodology not only avoids the data-snooping biases that are inherent in the portfolio based approaches (see Lo and MacKinlay (1990)) but also avoids the error-in-variables bias created by errors in estimating factor loadings. The use of individual stocks as test assets is robust to the sensitivity of asset pricing tests to the portfolio grouping procedure.

We assume that returns are generated by a K -factor model

$$R_{jt} = E_{t-1}(R_{jt}) + \sum_{k=1}^K \beta_{jkt-1} f_{kt} + e_{jt}, \quad (2)$$

where E_{t-1} is the conditional expectations operator, R_{jt} is the return on security j at time t , f_{kt} is the unanticipated (with respect to information available at $t-1$) time t return on the k 'th factor, and β_{jkt-1} is the conditional beta corresponding to the k 'th factor. $E_{t-1}(R_{jt})$ is modeled using the exact pricing specification

$$E_{t-1}(R_{jt}) - R_{Ft} = \sum_{k=1}^K \lambda_{kt-1} \beta_{jkt-1}, \quad (3)$$

where R_{Ft} is the riskfree rate and λ_{kt} is the risk premium for factor k at time t . The estimated risk-adjusted return on each security for month t is then calculated as:

$$R_{jt}^* \equiv R_{jt} - R_{Ft} - \sum_{k=1}^K \hat{\beta}_{jkt-1} F_{kt}, \quad (4)$$

where $F_{kt} \equiv f_{kt} + \lambda_{kt}$ is the sum of the factor innovation and its corresponding risk premium and $\hat{\beta}_{jkt}$ is the beta estimated by a first-pass time-series regression over the entire sample period.¹² Our risk adjustment procedure imposes the assumptions that the conditional zero-beta return equals the conditional risk-free rate, and that the factor

¹²While this entails the use of future data in calculating the factor loadings, Fama and French (1992) have shown that this forward looking does not impact any of the results.

premium is equal to the excess return on the factor, as is the case when factors are portfolio based. We use the Fama-French (1993) factors.

Next, we run the cross-sectional regression

$$R_{jt}^* = c_{0t} + \sum_{m=1}^M c_{mt} Z_{mjt-1} + e_{jt}, \quad (5)$$

where Z_{mjt-1} is the value of characteristic m for security j at time $t - 1$, and M is the total number of characteristics. Under exact pricing, equity characteristics do not explain risk-adjusted return, and under the null should be insignificant in the specification (5). To examine significance, we estimate the vector of characteristics rewards each month as

$$\hat{c}_t = (Z'_{t-1} Z_{t-1})^{-1} Z'_{t-1} R_t^*, \quad (6)$$

where Z_{t-1} is a matrix including the M firm characteristics for N_t test assets and R_t^* is the vector of risk-adjusted returns on all test assets. The standard Fama-MacBeth (1973) estimators are the time-series averages of these coefficients, $\hat{c}_t$. The standard errors of the estimators are obtained from the time series of monthly estimates.

We require a firm to satisfy the following criteria in order to be included for analysis in a given month: (1) Its return in the current month, t , be available from CRSP, and sufficient data be available to calculate the size, price, and dividend yield as of month $t-2$; (2) Sufficient data be available on the COMPUSTAT tapes to calculate the book to market ratio as of December of the previous year. We use a number of firm-specific characteristics as controls following Brennan et al. (1998). For each stock the following variables were calculated each month:

- SIZE: the natural logarithm of the market value of the equity of the firm as of the end of the second to last month.
- BM: the natural logarithm of the ratio of the book value of equity plus deferred taxes to the market value of equity, using the end of the previous year market and book values. As in Fama and French (1992), the value of BM for July of year t to June of year $t+1$ was computed using accounting data at the end of year $t-1$, and book-to-market ratio values greater than the 0.995 fractile or less than the 0.005 fractile were set equal to the 0.995 and 0.005 fractile values, respectively.

- DVOL: the natural logarithm of the dollar trading volume in the second to last month.
- SUE: the most recent standardized unexpected earnings.
- RET12: the cumulative return over the 12 months ending two months previously.

The results are presented in Table 7. The first regression, essentially repeats earlier results in BCS and Avramov and Chordia (2005). Firm characteristics do matter. Small stocks, high book-to-market stocks, stocks with low dollar trading volume and stocks with high past twelve month returns all have higher risk adjusted returns. The second regression adds the firm characteristic SUE as an additional independent variable. The coefficient on SUE is positive and highly significant suggesting that stocks with high earnings surprises have higher returns. With SUE as one of the dependent variables the coefficient on the past twelve month returns declines and becomes statistically insignificant at the 5% level. This suggests that earnings and price momentum are related and is consistent with the results in Chordia and Shivakumar (2005).

The innovation in this paper is the use of interaction terms. The third regression introduces an interaction term of firm size and SUE. The coefficient on the interaction term is significantly negative suggesting that small firms exhibit more earnings momentum than large firms.¹³ In the fourth regression we introduce an interaction term of SUE and the Amihud illiquidity measure. The coefficient on this interaction term is significantly positive suggesting that the impact of SUE on risk adjusted returns is higher in more illiquid stocks. The last regression uses both interaction terms as dependent variables. The coefficient estimates on both interaction terms become statistically insignificant possibly because firm size and illiquidity are (negatively) correlated. In other words, both size and illiquidity are equally important conditioning variables for SUE in its relation with risk adjusted returns of individual stocks.

Overall, both firm size and liquidity are important drivers of the post-earnings-announcement-drift. The drift is stronger in small, illiquid stocks.

¹³We have also used an interaction term of stock price and SUE. While negative, this interaction term was statistically insignificant.

4 Transaction Costs

Table 2 shows that the potential profits from a strategy that goes long the high SUE portfolio and short the low SUE portfolio can be substantial, specially for the low liquidity stocks. Given that the potential profits obtain amongst the illiquid stocks, we suspect that transaction costs may render any trading strategy unprofitable. This suspicion is confirmed when examining the effective spreads in Table 3. The proportional bid ask spread of the highly illiquid stocks with the lowest (highest) SUE is 1.88% (1.80%). Thus, the total cost of setting up a long-short position is 3.68% and the round trip cost is 7.36%. The potential profit is 1.79% per month or 10.7% over six months. Thus, the bid-ask spreads account for over 67% of the potential profits of the post-earnings-announcement-drift. However, institutional investors carefully watch transactions costs. They choose trade size and spread out their trades over days so as to minimize trading costs. Therefore, the proportional spread of 3.68% may be an overstatement and a more precise estimation of the trading costs is warranted.

We now explicitly examine institutional transaction costs. Specifically, we use the institutional transaction cost estimates of Keim and Madhavan (1997) (henceforth KM) within a dynamic framework to estimate net returns based on a strategy designed to exploit the drift. In addition, we also consider the market impact cost analysis of Korajczyk and Sadka (2004) and Chen, Stanzl and Watanabe (2004).

4.1 KM Estimates of Transaction Costs

KM provide estimates of market impact as well as commission costs for buy and sell trades for NYSE-AMEX stocks (as well as NASDAQ stocks) traded by 21 large institutions during 1991 to 1993. Their transaction cost data is categorized by trade size as well as by firm size. We will focus on the smallest trade sizes to minimize trading costs. We will also use the various size classifications from Table 3. We focus on the most illiquid stocks because the potential profits are the highest for these portfolios. The strategy of going long the high SUE stocks and short the low SUE stocks results in monthly returns of 1.79% (see Table 2). The stocks in this portfolio belong to the smallest size quintile. According to KM, the cost of a small buy trade for NYSE-AMEX stocks in the smallest size quintile is 0.39%, and the cost of a small sell trade is 0.75% for a total cost of 1.14%. Thus, after accounting for transactions costs the long-short portfolio return is 0.65% per month.

The caveat with the above analysis is that the data in KM pertains to institutional trades during 1991-1993. While, the above costs includes commissions as well as market impact costs, KM show that there is large variation in trading costs across institutions, investment styles, and trade difficulty. Trading costs are likely to be higher during the early part of the sample. Using revenues from broker and dealer firms to estimate trading costs, Stoll (1995) finds that costs have declined by over 40% over 1980 to 1992. We will now conduct a dynamic analysis of the trading costs by using the methodology in Cooper, Gutierrez, and Marcum (2005). This methodology applies the trading costs estimates from Stoll (1995) to the sample specific estimates of KM. The idea is adjust the cost estimates of KM for intertemporal variation using the cost estimates from Stoll (1995).¹⁴ In order to be conservative, we will (i) estimate costs for value traders since they have the lowest trading costs, (ii) set the trade size to zero, and (iii) place a cap of 2.5% on the trading costs for each stock. The results are presented in Table 5, Panel A.

Consider the portfolio which has the most illiquid stocks and the lowest SUE. From Table 2, the gross average monthly return of this portfolio is 0.25%. After adjusting for the Keim and Madhavan cost estimates of 0.63%, the net monthly return is -0.38%. Similarly, the most illiquid, high SUE portfolio has a gross average monthly return of 2.04% and with costs estimates of 0.59%, the net return is 1.45%. The raw returns for the high SUE minus the low SUE portfolio decline from 1.79% in Table 2 to 0.56% after adjusting for transactions costs of 1.22% (0.63%+0.59%). Transactions costs amount to about 67% of the potential profits. The potential profits, in Table 2, from the long short SUE portfolios ranged from 0.24% to 1.79%. After accounting for transactions costs the profits range from 0.21% to 0.56% per month. Thus, while the paper profits seem large, a large fraction of these profits are not realizable.

While it seems that the above transaction costs are large, recall that we have been quite conservative in estimating these costs. The costs may actually be understated even more due to the following reasons:

1. KM use data for trades that were actually consummated. If some trades were abandoned due to high transactions costs then the KM estimates of trading costs would be biased downwards.
2. The transaction cost estimates in KM do not include short selling costs.
3. Since the transactions costs have been estimated for a zero trade size, the above

¹⁴For details, see the appendix in Cooper, Gutierrez, and Marcum (2005).

analysis has abstracted from market impact costs. For any meaningful trade size there will be substantial market impact costs.

To provide yet another dimension of transaction costs that accounts for market impact costs, we next implement the analysis of Korajczyk and Sadka (2004).

4.2 Korajczyk and Sadka Estimates of Market Impact Costs

Korajczyk and Sadka (2004) use intraday transactions data to estimate market impact costs. These costs are calculated each month for each stock over the period January 1993 through May 1997. Using cross-sectional relationships between these market impact costs and firm characteristics, the costs are then estimated for the entire sample period. The estimated market impact costs are then used to compute the cost of the taking positions in the different illiquidity-SUE portfolios each month.¹⁵ The results are presented for five million dollars of investment each month. In 1972, this is equivalent to about 23 million in 2004 dollars.

Panel B of Table 5 shows that upon adjusting for market impact costs the strategy of going long the high SUE stocks and short the low SUE stocks results in net returns that are significantly lower than the gross returns. For the highest illiquidity portfolio, the raw returns from a long-short strategy are -58% per month upon accounting for market impact costs. For the second highest illiquidity portfolio the corresponding return while positive is only 0.37%. Thus, an investment portfolio as small as \$5,000,000 has enough market impact costs to eliminate the potential profits from a strategy that exploits the post-earnings-announcement-drift. Even so, these costs understate the true transactions costs because they do not account for commissions or short sale costs. In Table 2 of their paper, KM show that commissions for institutional investors can be as high as 60% of the trading costs.

In the next subsection, we consider yet another measure of market impact costs as proposed by Chen, Stanzl and Watanabe (2004).

¹⁵For details, see Appendix A in Korajczyk and Sadka (2004).

4.3 Chen, Stanzl and Watanabe Estimates of Market Impact Costs

Chen, Stanzl and Watanabe (2004) also use transactions data to estimate market impact costs. However, they allow for a non-linear price impact function. In their model, trading volume non-linearly impacts price changes as measured by the mid-point of the bid-ask quote. They provide parameter estimates of the model by size decile and we use these estimates to measure the market impact costs for a five million dollar investment each month. The results in Panel C of Table 5 suggest that market impact costs have a big impact on the net returns from the long-short strategy. The average net returns are, once again, either negligible or negative. For the most illiquid stocks, the net return is -0.20% per month. The highest net return is an economically insignificant 0.17% per month. Once again, as was the case with Korajczyk and Sadka (2004), the market impact cost measures of Chen, Stanzl and Watanabe (2004) may actually understate the total trading costs because they do not account for commissions and for short selling costs.

4.4 Subsample Analysis

Trading costs have declined over time. Chordia, Roll and Subrahmanyam (2001) have documented a steady decrease in bid-ask spreads over time. In this section, we ask whether the long-short trading strategies have become more profitable over time. We split our sample into two periods: 1972-1987 and 1988-2003. Table 6 presents the results for the long-short portfolios. Costs have indeed declined substantially over the two subperiods. For instance, during 1972-1987, the KM trading costs for a long-short position range from 0.05% per month for the liquid stocks to 1.75% per month for the most illiquid stocks. During 1988-2003, the KM trading costs are lower and they range from 0.01% for the liquid stocks to 0.74% per month for the most illiquid stocks. A similar pattern exists for the Korajczyk and Sadka (2004) costs as well as for the Chen, Stanzl and Watanabe (2004) estimates of the market impact costs. For the most illiquid stocks, the Korajczyk and Sadka (Chen, Stanzl and Watanabe) estimates of costs declined from 3.47% (2.68%) during 1972-1987 to 1.72% (1.36%) during 1988-2003.

While the costs have declined over time, the potential profitability of the post-earnings-announcement-drift has also declined over time. During 1972-1987, the gross returns ranged from 0.30% for the liquid stocks to 1.84% for the most illiquid stocks.

During 1988-2003, the corresponding returns ranged from 0.18% to 1.74% per month. The risk-adjusted returns have also declined over time. For the most illiquid stocks, the alpha from the Fama-French model has declined from 2.05% per month during 1972-1987 to 1.74% per month during 1988-2003.

After controlling for transactions costs the net profitability is actually higher during the latter half of the sample period. For the most illiquid stocks, the Fama-French alpha, after adjusting for KM transactions costs,¹⁶ is 0.09% per month during 1972-1987 but it is 1.01% per month during 1988-2003. The net return after adjusting for Korajczyk and Sadka (2004) estimates of trading costs is -1.62% per month in the first half of the sample and 0.02% per month during the second half of the sample.

The overall conclusion is that while the potential returns from the long-short strategy has declined over time, trading costs have declined even more. Thus, the potentially profits from the long-short strategy are higher during the second half of the sample. Even so, after adjusting for transactions costs including market impact costs, the potential profits are not realizable.

Figure 1 presents five year moving averages of the gross returns from the long-short strategy for the most illiquid stocks as well as the transactions costs obtained using the analysis of (i) Keim and Madhavan (1997), (ii) Korajczyk and Sadka (2004) for a five million dollar portfolio, and (iii) Chen, Stanzl and Watanabe (2004) also for a five million dollar portfolio. The potential profits as measured by gross returns are quite variable. These profits were high in the early nineties and also in the period 2000-2004. The Korajczyk and Sadka transactions costs were estimated using transactions data over the period 1993-1997 and have been estimated out of sample, from 1972-1992 and 1993-2004, by using a cross-sectional relation between the costs and firm characteristics. Since the firm characteristics are quite variable the Korajczyk and Sadka estimate of transactions costs are also quite variable. The Keim and Madhavan as well as the Chen, Stanzl and Watanabe estimate of costs are not as variable because we only have these estimates for different size groups over a short time interval. By design, all the cost estimates decrease over time in line with the pattern observed by Stoll (1995).

The transaction cost estimates obtained on the basis of Chen, Stanzl and Watanabe (2004) are consistently lower than those obtained using Korajczyk and Sadka (2004), even though both methods capture market impact costs. The reason for this goes back to the fact that we have cost estimates for each individual stock from Korajczyk and

¹⁶Recall that the Keim and Madhavan costs do not account for market impact.

Sadka (2004) but only the estimates for size deciles from Chen, Stanzl and Watanabe (2004). Within the smallest size decile, the average cost estimate is likely to understate the costs for the smallest stocks when the Chen, Stanzl and Watanabe (2004) estimates are used. The Korajczyk and Sadka (2004) estimate of transactions costs are likely to be more accurate because these estimates are available at the individual stock level.

Focusing on the Korajczyk and Sadka (2004) estimate of transactions costs we see that the potential profits, from a long-short position in high SUE stocks and low SUE stocks respectively, are lower than the cost estimates for almost the entire period. The only exceptions are the periods 1990-1995 and 2000-2004. During the period 1990-1995, the potential profits are indeed higher than the costs but only marginally so. It is only in the period 2000 through 2004 that the potential profits are as much as a hundred basis points higher than the costs. Although at the end of 2004, the difference between the potential profits and transactions costs is down to 20 basis points. The high realizable profits in 2000-2004 are likely to disappear if commission costs and short sale costs are accounted for and as the size of the portfolio increases above five million.

4.5 Summary of Transactions Costs

This paper analyzes trading costs from a number of different perspectives, including proportional effective bid-ask spreads, dynamic institutional trading costs, and the market impact costs. Consider the most illiquid stocks. The proportional effective bid-ask spread cost for the long-short portfolio calculated over the sample period 1988 through 2002 is 3.68%. The dynamic cost analysis of institutional trading costs using the methodology of Cooper, Gutierrez, and Marcum (2005) suggests that transactions costs including commissions (but excluding market impact costs) are 1.21%. The market impact costs (not including commission costs) using the Korajczyk and Sadka (2004) methodology amount to 2.37% for an investment portfolio of \$5,000,000. Finally, the market impact costs using the Chen, Stanzl and Watanabe estimates amounts to 1.99%. These costs eliminate the potential profits from a long-short strategy that exploits the post-earnings-announcement drift.

The overall conclusion from an analysis of bid-ask spreads, market impact costs, and institutional transactions costs is that most of the net profits disappear upon accounting for trading costs. The transactions costs account for 66% to 100% of the potential profits from the earnings momentum strategy. The main reason is that the largest potential profits are obtained in the highly illiquid stocks that have high trading costs.

5 Earnings Announcements

It has been documented that a large fraction of the post-earnings-announcement-drift occurs at the subsequent earnings announcements. For instance, Chan, Jegadeesh and Lakonishok (1996) document that of the 6.8% return to the long-short portfolio over the six month formation period, 3.2% obtains around the earnings announcement dates over the same six month formation period. In other words, 47% of the drift is realized around the subsequent earnings announcement dates. The returns around earnings announcements are measured over a three day period from $t - 2$ to $t + 1$ with t being the earnings announcement date. Since we know that trading volumes are high around earnings announcements, it might be the case that liquidity is higher and transactions costs are lower. With lower transactions costs it is possible that the long-short portfolio strategy becomes profitable.

In this section, we examine liquidity around earnings announcement dates. More specifically, we examine bid-ask spreads and quoted depths for NYSE stocks around earnings announcement dates from $t - 2$ to $t + 1$, over the period 1988-2002. Our daily liquidity measures over $t - 2$ to $t + 1$ are computed relative to the same measures over $t - 10$ to $t - 6$ and $t + 6$ to $t + 10$. Table 8 reports the difference between the daily average liquidity measures around earnings announcement ($t - 2$ to $t + 1$) and the daily average over periods away from the earnings announcement ($t - 10$ to $t - 6$ and $t + 6$ to $t + 10$). The results are grouped by illiquidity.

Trading volume increases around earnings announcements for all stocks regardless of the level of illiquidity. For the most liquid stocks, the daily trading volume increases by an average of 368,738 shares but for the most illiquid stocks the increase is only 2,305 shares. The result for the quoted depth is mixed. It increases for some of the illiquidity deciles and decreases for others. The overall daily change in depth is a decrease of about 180 shares. For the most illiquid stocks, depth increases by a negligible 19 shares. The proportional quoted spread increases for all stocks except for the most illiquid stocks where it decreases by 0.034%. The proportional effective spread, which arguably is a more reliable measure of actual trading costs, increase across the board. For the most illiquid stocks, the proportional effective spread increases by 0.024%.

Overall, liquidity decreases around earnings announcement dates. Thus, transactions costs are unlikely to be lower around earnings announcements and it is not likely that the long-short portfolio strategy becomes profitable due to lower transactions costs.

6 Conclusions

The post-earnings-announcement-drift is one of the most robust anomalies that was first documented by Ball and Brown (1968). A portfolio that is long the high earnings surprise stocks and short the low earnings surprise stocks earns a return of 90 basis points per month. A long-standing, robust anomaly that can be exploited using a simple long-short strategy points to a violation of weak form efficiency. What is specially intriguing is that stock prices do not respond completely and immediately to information as visible and freely available as earnings announcements. Moreover, the earnings momentum anomaly has persisted for almost four decades.

This paper shows that transactions costs inhibit the exploitation of the post-earnings-announcement-drift thereby causing it to seem to be robust over four decades. Earnings momentum profits accrue mainly in the highly illiquid stocks. These illiquid stocks have high trading costs and high market impact costs. Transaction costs account for 66% to 100% of the potential paper profits from the long-short strategy designed to exploit the earnings drift. The market impact costs of a long-short strategy are such that a five million dollar mutual fund designed to exploit the post-earnings-announcement-drift is no longer profitable. This lack of profitability suggests that the violation of the efficient market hypothesis due to earnings momentum is not so egregious after all.

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Table 1: SUE-Sorted Portfolios

This table presents mean SUE, Illiquidity, and returns of SUE-sorted portfolios. SUE is measured as $\frac{\text{SUE}}{\text{SUE}_{10-1}}$. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume multiplied by 1×10^6). The breakpoints for sorts are determined by the last three months SUE. The portfolios are formed every month and are held for six months - the time series of portfolio returns is based on equal-weighted average of six portfolios each month. Returns are given in percent per month. The sample period is 1972 to 2004 and the portfolios include all NYSE and AMEX stocks with price greater than five dollars. The table also reports alphas (in percent per month) for the 10-1 portfolio. Alphas are calculated from a market model, a three-factor model based on Fama and French (1993), and a four-factor model that also includes a momentum factor.

Portfolio	SUE	Illiquidity	Return
1	-3.740	0.638	0.882
2	-1.168	0.705	1.010
3	-0.573	0.733	1.061
4	-0.205	0.729	1.163
5	0.044	0.684	1.237
6	0.255	0.654	1.321
7	0.492	0.625	1.409
8	0.825	0.592	1.485
9	1.397	0.583	1.524
10	3.216	0.431	1.617
10-1	6.956	0.535	0.734
Mkt Alpha			0.74 (7.79)
FF3 Alpha			0.87 (7.44)
FF3+UMD Alpha			0.65 (5.71)

Table 2: Gross Returns of SUE- and Illiquidity-Sorted Portfolios

This table presents mean gross returns of SUE- and Illiquidity-sorted portfolios. SUE is measured as $\frac{\text{Amihud}}{\text{SUE}}$. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume multiplied by 1×10^6). The breakpoints for sorts are determined by the last three months SUE and the last month Amihud measure. SUE (illiquidity) portfolio 1 has the lowest SUE (illiquidity), and portfolio 10 has the highest SUE (illiquidity). The stocks are sorted independently based on SUE and illiquidity. The portfolios are formed every month and are held for six months - the time series of portfolio returns is based on equal-weighted average of six portfolios each month. Returns are given in percent per month. The sample period is 1972 to 2004 and the portfolios include all NYSE and AMEX stocks with price greater than five dollars. The table also reports alphas (in percent per month) for the 10–1 SUE portfolio. Alphas are calculated from a market model, a three-factor model based on Fama and French (1993), a four-factor model that includes a momentum factor, a five-factor model that includes a liquidity factor based on Pastor and Stambaugh (2003).

SUE Rank	Illiquidity Rank									
	1	2	3	4	5	6	7	8	9	10
Gross Returns										
1	0.99	1.14	1.11	1.08	0.98	0.80	0.94	0.64	0.52	0.25
2	0.87	1.01	1.09	1.04	1.11	1.09	1.17	0.96	1.04	0.68
3	0.85	1.11	1.10	1.07	1.10	1.05	1.18	1.22	1.04	0.88
4	0.98	1.01	1.07	1.10	1.33	1.20	1.28	1.21	1.14	1.27
5	0.97	1.15	1.22	1.28	1.15	1.17	1.29	1.27	1.39	1.46
6	1.05	1.10	1.26	1.20	1.24	1.36	1.43	1.56	1.57	1.46
7	1.07	1.16	1.28	1.27	1.31	1.53	1.52	1.66	1.70	1.83
8	1.10	1.17	1.34	1.34	1.44	1.54	1.62	1.86	1.80	1.89
9	1.11	1.19	1.18	1.36	1.56	1.59	1.68	1.80	1.93	2.16
10	1.23	1.43	1.50	1.51	1.70	1.64	1.86	1.92	1.99	2.04
10–1	0.24	0.29	0.39	0.43	0.72	0.83	0.91	1.28	1.47	1.79
Alphas for 10–1 Portfolio										
Mkt	0.26 (2.10)	0.32 (2.29)	0.41 (2.59)	0.44 (2.71)	0.72 (4.83)	0.83 (5.19)	0.95 (6.72)	1.28 (8.48)	1.44 (9.89)	1.74 (9.45)
FF3	0.33 (2.51)	0.46 (2.91)	0.51 (2.80)	0.59 (3.23)	0.83 (5.36)	0.93 (5.03)	0.94 (5.77)	1.37 (8.39)	1.51 (9.96)	1.86 (9.57)
FF3+UMD	0.16 (1.21)	0.19 (1.28)	0.21 (1.16)	0.37 (2.11)	0.63 (4.21)	0.78 (4.19)	0.74 (4.27)	1.22 (6.66)	1.31 (7.85)	1.74 (8.40)
FF3+UMD+PS	0.19 (1.53)	0.39 (2.16)	0.51 (2.72)	0.56 (2.71)	0.93 (5.66)	1.02 (5.74)	0.96 (5.66)	1.26 (6.97)	1.51 (9.46)	1.81 (8.37)

Table 3: Descriptives of SUE- and Illiquidity-Sorted Portfolios

This table presents portfolio characteristics of SUE- and Illiquidity-sorted portfolios. SUE is measured as ???. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume multiplied by 1×10^6). The breakpoints for sorts are determined by the last three months SUE and the last month Amihud measure. SUE (illiquidity) portfolio 1 has the lowest SUE (illiquidity), and portfolio 10 has the highest SUE (illiquidity). The stocks are sorted independently based on SUE and illiquidity. The portfolios are formed every month and are held for six months - the time series of portfolio returns is based on equal-weighted average of six portfolios each month. Portfolio ‘Sz Rank’ is the average size rank of the stocks in the portfolio, where the size rank (from 1 to 10) is determined from NYSE breakpoints. Portfolio ‘Illiquidity’ is computed using the Amihud measure and is expressed as daily illiquidity (multiplied by 10^6). Portfolio ‘Turnover’ is daily turnover in percent, and ‘DVolume’ is dollar volume in millions of dollars. Portfolio ‘PESPR’ is the proportional effective spread, ‘PQSPR’ is the proportional quoted spread, and ‘Depth’ is the average depth. Depth is in thousands of shares and proportional spreads are in percent. The sample period in Panel A is 1972 to 2004 and the portfolios include all NYSE and AMEX stocks with price greater than five dollars. The sample period in Panel B is 1988 to 2004 and the portfolios include all NYSE stocks with price greater than five dollars.

Panel A: NYSE and AMEX Sample for 1972 to 2004

SUE Rank		Illiquidity Rank									
		1	2	3	4	5	6	7	8	9	10
1	SzRank	9.621	8.373	7.149	5.973	4.918	3.881	3.095	2.325	1.728	1.667
	Illiquidity	0.005	0.015	0.035	0.075	0.135	0.237	0.452	0.787	1.576	4.651
	Turnover	0.697	0.762	0.760	0.721	0.666	0.602	0.485	0.396	0.289	0.194
	DVolume	8.789	2.231	1.054	0.565	0.328	0.184	0.103	0.047	0.019	0.006
2	SzRank	9.615	8.374	7.199	6.094	5.000	4.049	3.193	2.421	1.764	1.560
	Illiquidity	0.005	0.016	0.035	0.071	0.135	0.229	0.412	0.762	1.514	4.466
	Turnover	0.703	0.746	0.734	0.660	0.605	0.533	0.443	0.377	0.277	0.188
	DVolume	9.178	2.219	1.051	0.563	0.318	0.183	0.099	0.049	0.021	0.009
3	SzRank	9.541	8.343	7.122	5.985	4.925	4.037	3.153	2.430	1.756	1.604
	Illiquidity	0.005	0.015	0.035	0.070	0.140	0.235	0.402	0.750	1.528	4.503
	Turnover	0.740	0.762	0.751	0.695	0.625	0.541	0.456	0.384	0.280	0.191
	DVolume	8.794	2.204	1.081	0.577	0.322	0.182	0.099	0.056	0.020	0.006
4	SzRank	9.552	8.322	7.151	6.010	5.008	3.986	3.130	2.401	1.797	1.655
	Illiquidity	0.005	0.016	0.035	0.074	0.127	0.223	0.406	0.732	1.518	4.374
	Turnover	0.751	0.757	0.720	0.668	0.608	0.545	0.456	0.369	0.273	0.199
	DVolume	8.683	2.222	1.055	0.563	0.326	0.185	0.097	0.053	0.020	0.006
5	SzRank	9.592	8.352	7.160	6.056	4.999	4.014	3.112	2.381	1.712	1.653
	Illiquidity	0.006	0.016	0.035	0.070	0.125	0.227	0.413	0.724	1.443	4.183
	Turnover	0.699	0.722	0.718	0.657	0.610	0.541	0.460	0.384	0.292	0.192
	DVolume	8.375	2.102	1.021	0.561	0.334	0.185	0.107	0.055	0.021	0.007
6	SzRank	9.646	8.413	7.224	6.061	4.996	4.030	3.217	2.404	1.740	1.678
	Illiquidity	0.004	0.015	0.034	0.071	0.128	0.234	0.399	0.710	1.453	4.321
	Turnover	0.675	0.720	0.695	0.643	0.614	0.532	0.451	0.383	0.301	0.208
	DVolume	8.812	2.087	1.029	0.559	0.334	0.185	0.103	0.054	0.022	0.006
7	SzRank	9.636	8.421	7.181	6.060	4.961	4.046	3.168	2.401	1.742	1.652
	Illiquidity	0.007	0.015	0.034	0.067	0.127	0.218	0.389	0.704	1.448	4.258
	Turnover	0.673	0.710	0.733	0.665	0.614	0.546	0.457	0.401	0.304	0.205
	DVolume	8.776	2.128	1.030	0.563	0.327	0.187	0.103	0.053	0.021	0.007
8	SzRank	9.623	8.355	7.195	6.035	5.022	4.070	3.197	2.401	1.755	1.575
	Illiquidity	0.005	0.015	0.033	0.068	0.123	0.215	0.382	0.692	1.411	4.149
	Turnover	0.734	0.765	0.729	0.694	0.604	0.547	0.477	0.391	0.305	0.206
	DVolume	9.005	2.154	1.049	0.569	0.326	0.189	0.104	0.052	0.022	0.006
9	SzRank	9.609	8.291	7.126	5.978	4.930	4.021	3.149	2.388	1.736	1.517
	Illiquidity	0.004	0.015	0.034	0.065	0.122	0.220	0.383	0.682	1.380	3.954
	Turnover	0.735	0.800	0.762	0.704	0.644	0.558	0.474	0.396	0.308	0.220
	DVolume	8.495	2.093	1.055	0.564	0.335	0.191	0.103	0.053	0.020	0.008
10	SzRank	9.603	8.294	7.140	5.953	4.930	3.987	3.148	2.356	1.734	1.507

Panel B: NYSE Sample for 1988 to 2004

SUE Rank		Illiquidity Rank									
		1	2	3	4	5	6	7	8	9	10
1	PESPR	0.244	0.350	0.462	0.540	0.667	0.833	0.977	1.197	1.449	1.880
	PQSPR	0.361	0.526	0.686	0.806	0.996	1.240	1.459	1.810	2.199	2.801
	Depth	9.657	7.290	6.363	5.208	4.586	3.904	3.092	2.678	2.304	1.937
2	PESPR	0.246	0.341	0.437	0.512	0.630	0.761	0.924	1.128	1.399	1.923
	PQSPR	0.363	0.516	0.657	0.770	0.944	1.146	1.394	1.701	2.135	2.928
	Depth	9.582	6.817	5.745	4.607	4.055	3.400	2.755	2.373	2.257	1.837
3	PESPR	0.251	0.345	0.439	0.528	0.627	0.763	0.911	1.131	1.439	1.918
	PQSPR	0.370	0.520	0.657	0.786	0.942	1.144	1.379	1.712	2.180	2.855
	Depth	9.237	7.500	6.346	4.934	4.058	3.271	2.843	2.637	2.166	1.907
4	PESPR	0.256	0.351	0.447	0.517	0.633	0.776	0.945	1.155	1.439	1.877
	PQSPR	0.376	0.522	0.664	0.773	0.946	1.162	1.425	1.746	2.191	2.837
	Depth	10.096	7.799	6.582	5.161	4.439	3.663	3.178	2.606	2.251	1.902
5	PESPR	0.255	0.347	0.435	0.527	0.620	0.770	0.939	1.161	1.443	1.823
	PQSPR	0.374	0.519	0.651	0.786	0.927	1.151	1.410	1.750	2.184	2.770
	Depth	10.231	7.488	6.234	5.618	4.406	3.825	3.218	2.727	2.296	1.977
6	PESPR	0.229	0.322	0.415	0.499	0.599	0.742	0.870	1.105	1.401	1.825
	PQSPR	0.341	0.489	0.627	0.751	0.901	1.121	1.322	1.677	2.145	2.803
	Depth	9.042	6.768	5.943	4.984	4.061	3.499	2.784	2.496	2.156	1.962
7	PESPR	0.223	0.317	0.399	0.475	0.586	0.697	0.859	1.053	1.319	1.824
	PQSPR	0.331	0.482	0.607	0.726	0.892	1.059	1.296	1.588	2.019	2.706
	Depth	8.393	6.252	5.094	4.248	3.742	3.058	2.670	2.413	1.990	1.869
8	PESPR	0.219	0.317	0.389	0.482	0.576	0.680	0.821	1.000	1.291	1.742
	PQSPR	0.328	0.483	0.596	0.737	0.877	1.041	1.254	1.519	1.957	2.647
	Depth	7.635	6.198	4.938	4.183	3.377	2.934	2.458	2.262	1.911	1.770
9	PESPR	0.221	0.307	0.399	0.469	0.573	0.691	0.834	1.002	1.278	1.841
	PQSPR	0.334	0.470	0.607	0.713	0.873	1.050	1.272	1.533	1.952	2.790
	Depth	8.306	5.537	5.187	3.947	3.432	2.957	2.493	2.261	1.894	1.799
10	PESPR	0.213	0.302	0.385	0.462	0.552	0.703	0.837	1.022	1.278	1.796
	PQSPR	0.323	0.462	0.584	0.707	0.842	1.064	1.275	1.551	1.964	2.765
	Depth	7.013	5.362	4.787	3.936	3.304	2.960	2.518	2.228	1.964	1.818

Table 4: Size or Illiquidity

This table presents mean SUE, Illiquidity, and returns of SUE, Illiquidity, and size sorted portfolios. SUE is measured as $\frac{R_{it}}{P_{it-1}}$. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume multiplied by 1×10^6). The breakpoints for sorts are determined by the last three months SUE, the last month Amihud measure, and last month size. Portfolio 1 has the lowest SUE/illiquidity/size, and portfolio 4 has the highest SUE/illiquidity/size. The stocks are sorted first sorted on size, then on illiquidity, and finally on SUE in panel A while they are sorted first sorted on illiquidity, then on size, and finally on SUE in panel B. The portfolios are formed every month and are held for six months - the time series of portfolio returns is based on equal-weighted average of six portfolios each month. Returns are given in percent per month. The sample period is 1972 to 2004 and the portfolios include all NYSE and AMEX stocks with price greater than five dollars.

Panel A: Conditional on Size								
SUE	Size Rank = 1 Illiquidity Rank				Size Rank = 4 Illiquidity Rank			
	1	2	3	4	1	2	3	4
1	0.748	0.807	0.688	0.433	0.876	0.969	1.065	1.160
2	1.021	1.206	1.152	1.225	0.899	1.025	1.127	1.046
3	1.526	1.496	1.622	1.628	1.046	1.122	1.175	1.242
4	1.854	1.952	1.949	2.010	1.141	1.242	1.286	1.318
4-1	1.106	1.145	1.262	1.577	0.265	0.273	0.221	0.157
Panel B: Conditional on Illiquidity								
SUE	Illiquidity Rank = 1 Size Rank				Illiquidity Rank = 4 Size Rank			
	1	2	3	4	1	2	3	4
1	1.099	0.998	1.023	0.951	0.560	0.593	0.866	0.903
2	1.098	1.114	1.086	0.899	1.259	1.061	1.110	1.305
3	1.286	1.190	1.060	1.015	1.690	1.590	1.487	1.597
4	1.320	1.269	1.228	1.117	2.031	1.914	1.962	1.860
4-1	0.221	0.271	0.205	0.166	1.471	1.321	1.096	0.956

Table 5: Net Returns of SUE- and Illiquidity-Sorted Portfolios

This table presents mean net returns of SUE- and Illiquidity-sorted portfolios. SUE is measured as ???. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume multiplied by 1×10^6). The breakpoints for sorts are determined by the last three months SUE and the last month Amihud measure. SUE (illiquidity) portfolio 1 has the lowest SUE (illiquidity), and portfolio 10 has the highest SUE (illiquidity). The stocks are sorted independently based on SUE and illiquidity. The portfolios are formed every month and are held for six months - the time series of portfolio returns is based on equal-weighted average of six portfolios each month. Returns are given in percent per month. Net returns are calculated by subtracting Keim and Madhavan (1997), Korajczyk and Sadka (2004), and Chen, Stanzl, and Watanabe (2004) cost estimates. These are indicated by the mnemonics ‘KM’, ‘KS’, and ‘CSW’ respectively. The dollar amount of trading for KS and CSW estimates is five million dollars. The sample period is 1972 to 2004 and the portfolios include all NYSE and AMEX stocks with price greater than five dollars.

Panel A: Returns Net of KM Costs										
SUE Rank	Illiquidity Rank									
	1	2	3	4	5	6	7	8	9	10
1	0.97	1.08	1.01	0.92	0.77	0.53	0.61	0.24	0.04	-0.38
2	0.85	0.95	1.00	0.90	0.92	0.85	0.86	0.59	0.57	0.06
3	0.83	1.04	0.99	0.92	0.90	0.79	0.86	0.84	0.57	0.27
4	0.96	0.95	0.97	0.96	1.14	0.95	0.96	0.83	0.67	0.66
5	0.95	1.10	1.13	1.14	0.96	0.92	0.98	0.89	0.91	0.85
6	1.04	1.05	1.17	1.07	1.06	1.13	1.14	1.19	1.11	0.84
7	1.06	1.11	1.21	1.15	1.15	1.31	1.25	1.31	1.26	1.22
8	1.09	1.12	1.26	1.23	1.28	1.33	1.35	1.52	1.37	1.30
9	1.10	1.15	1.10	1.25	1.42	1.38	1.42	1.48	1.52	1.58
10	1.22	1.40	1.44	1.41	1.56	1.45	1.61	1.60	1.57	1.45
10-1	0.21	0.19	0.22	0.18	0.37	0.36	0.33	0.56	0.56	0.56

Panel B: Returns Net of KS Costs										
SUE Rank	Illiquidity Rank									
	1	2	3	4	5	6	7	8	9	10
1	0.87	0.97	0.89	0.82	0.69	0.48	0.60	0.23	-0.02	-0.82
2	0.75	0.83	0.88	0.80	0.84	0.79	0.84	0.58	0.56	-0.22
3	0.73	0.93	0.88	0.82	0.83	0.74	0.84	0.84	0.57	-0.02
4	0.86	0.83	0.85	0.86	1.06	0.90	0.95	0.86	0.69	0.43
5	0.86	0.99	1.01	1.04	0.88	0.88	0.97	0.90	0.94	0.64
6	0.95	0.93	1.06	0.97	0.98	1.08	1.12	1.21	1.10	0.51
7	0.96	1.00	1.09	1.04	1.06	1.25	1.22	1.30	1.23	0.85
8	0.99	1.00	1.14	1.12	1.19	1.27	1.31	1.52	1.32	0.82
9	1.01	1.03	0.98	1.14	1.32	1.32	1.38	1.45	1.46	1.00
10	1.14	1.29	1.32	1.30	1.47	1.37	1.56	1.55	1.43	0.74
10-1	0.04	-0.03	-0.01	-0.03	0.20	0.24	0.26	0.50	0.37	-0.58

Panel C: Returns Net of CSW Costs										
SUE Rank	Illiquidity Rank									
	1	2	3	4	5	6	7	8	9	10
1	0.88	0.97	0.91	0.84	0.70	0.45	0.49	0.00	-0.29	-0.67
2	0.76	0.84	0.88	0.80	0.83	0.76	0.72	0.35	0.23	-0.22
3	0.73	0.93	0.89	0.83	0.82	0.72	0.73	0.63	0.24	0.01
4	0.87	0.84	0.86	0.87	1.05	0.87	0.84	0.62	0.36	0.40
5	0.86	0.99	1.01	1.04	0.87	0.83	0.84	0.66	0.59	0.58
6	0.95	0.93	1.05	0.96	0.96	1.03	0.99	0.95	0.76	0.52
7	0.97	0.99	1.08	1.03	1.04	1.19	1.09	1.04	0.87	0.87
8	1.00	1.00	1.14	1.11	1.16	1.20	1.16	1.25	0.96	0.93
9	1.01	1.02	0.97	1.13	1.29	1.24	1.22	1.17	1.09	1.20
10	1.15	1.28	1.31	1.28	1.43	1.30	1.40	1.25	1.09	0.97
10-1	0.05	-0.03	-0.01	-0.03	0.17	0.14	-0.01	-0.02	-0.25	-0.20

Table 6: Returns of SUE- and Illiquidity-Sorted Portfolios: Sub-Sample Analysis

This table presents subs-sample analysis of returns of SUE- and Illiquidity-sorted portfolios. SUE is measured as $???$. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume multiplied by 1×10^6). The breakpoints for sorts are determined by the last three months SUE and the last month Amihud measure. SUE (illiquidity) portfolio 1 has the lowest SUE (illiquidity), and portfolio 10 has the highest SUE (illiquidity). The stocks are sorted independently based on SUE and illiquidity. The portfolios are formed every month and are held for six months - the time series of portfolio returns is based on equal-weighted average of six portfolios each month. Returns are given in percent per month. Net returns are calculated by subtracting Keim and Madhavan (1997), Korajczyk and Sadka (2004), and Chen, Stanzl, and Watanabe (2004) cost estimates. These are indicated by the mnemonics 'KM', 'KS', and 'CSW' respectively. The dollar amount of trading for KS and CSW estimates is five million dollars. The sample period is 1972 to 1987 in Panel A and 1988 to 2004 in Panel B and the portfolios include all NYSE and AMEX stocks with price greater than five dollars. The table also reports alphas (in percent per month) for the 10-1 SUE portfolio. Alphas are calculated from a three-factor model based on Fama and French (1993).

Panel A: 1972-1987

	1	2	3	4	5	6	7	8	9	10
	Illiquidity Rank									
Gross Return	0.30	0.60	0.66	0.63	1.05	1.06	1.15	1.28	1.79	1.84
FF3 alpha	0.41 (2.06)	0.85 (3.88)	0.87 (4.56)	0.87 (3.68)	1.19 (5.88)	1.25 (5.65)	1.33 (7.24)	1.53 (6.68)	1.91 (10.69)	2.05 (7.81)
Costs (KM)	0.05	0.17	0.28	0.41	0.56	0.70	0.85	1.04	1.31	1.75
Costs (KS)	0.30	0.46	0.56	0.64	0.71	0.81	0.88	1.07	1.56	3.47
Costs (CSW)	0.24	0.43	0.53	0.63	0.78	1.00	1.36	1.91	2.47	2.68
Net Return (KM)	0.25	0.43	0.39	0.23	0.50	0.35	0.30	0.23	0.48	0.09
Net Return (KS)	-0.00	0.14	0.11	-0.01	0.34	0.25	0.27	0.21	0.23	-1.62
Net Return (CSW)	0.06	0.17	0.13	0.00	0.27	0.05	-0.20	-0.63	-0.68	-0.84

Panel B: 1988–2004

	Illiquidity Rank									
	1	2	3	4	5	6	7	8	9	10
Gross Return	0.18	-0.00	0.13	0.25	0.41	0.63	0.70	1.28	1.17	1.74
FF3 alpha	0.24 (1.55)	0.13 (0.70)	0.26 (0.97)	0.38 (1.49)	0.57 (3.02)	0.73 (2.77)	0.68 (3.39)	1.32 (5.84)	1.21 (5.55)	1.74 (6.25)
Costs (KM)	0.01	0.02	0.07	0.11	0.17	0.26	0.33	0.43	0.54	0.74
Costs (KS)	0.12	0.19	0.26	0.32	0.36	0.42	0.46	0.55	0.73	1.72
Costs (CSW)	0.13	0.22	0.26	0.30	0.34	0.41	0.53	0.74	1.02	1.36
Net Return (KM)	0.18	-0.02	0.06	0.14	0.25	0.38	0.36	0.86	0.63	1.01
Net Return (KS)	0.07	-0.19	-0.13	-0.07	0.05	0.21	0.23	0.74	0.44	0.02
Net Return (CSW)	0.05	-0.22	-0.13	-0.06	0.07	0.22	0.17	0.55	0.14	0.38

Table 7: FamaMacbeth

This table presents the average β in the monthly cross-sectional Fama and Macbeth (1973) type regressions:

$$R_{it} - \beta_i F_{it} = \alpha_t + \beta_t X_{it-2} + \epsilon_{it}.$$

We calculate the β_i in a first pass regression using the entire sample period from Fama and French (1993) three-factor model. The characteristics X include: ‘Sz’ is the natural logarithm of the market value of equity, ‘BtoM’ is the ratio of book value of equity to market value of equity, ‘DVOL’ is the natural logarithm of the dollar volume of trading, ‘RET12’ is the cumulative return over the last twelve months, ‘SUE’ is the SUE from the last three months, and ‘Amihud’ is the ratio of absolute returns to dollar volume. All the characteristics are lagged by *two* months. t -statistics are in parenthesis below the coefficients are corrected for errors-in-variables following Shanken (1992). The sample period is 1972 to 2004 and the sample includes all NYSE and AMEX stocks.

	CNST	Sz	BtoM	DVol	Ret12	SUE	SUE*Sz	SUE*Ami
1.	2.173 (9.21)	-0.147 (-2.78)	0.184 (3.95)	-0.092 (-2.09)	0.279 (2.05)			
2.	2.133 (8.90)	-0.139 (-2.62)	0.192 (4.05)	-0.093 (-2.07)	0.211 (1.53)	0.053 (5.55)		
3.	2.136 (8.91)	-0.137 (-2.59)	0.190 (4.01)	-0.092 (-2.07)	0.187 (1.35)	0.185 (6.01)	-0.022 (-4.86)	
4.	2.174 (9.08)	-0.134 (-2.53)	0.193 (4.07)	-0.097 (-2.18)	0.190 (1.38)	0.116 (6.74)		0.018 (4.99)
5.	2.168 (9.00)	-0.137 (-2.57)	0.192 (4.05)	-0.095 (-2.10)	0.193 (1.39)	0.149 (2.92)	-0.010 (-0.77)	0.010 (0.99)

Table 8: Earnings Announcement Days

This table presents portfolio characteristics of SUE- and Illiquidity-sorted portfolios around earnings announcement dates. SUE is measured as $\frac{\Delta R}{\Delta P}$. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume multiplied by 1×10^6). The breakpoints for sorts are determined by the last three months SUE and the last month Amihud measure. SUE (illiquidity) portfolio 1 has the lowest SUE (illiquidity), and portfolio 10 has the highest SUE (illiquidity). The stocks are sorted independently based on SUE and illiquidity. The portfolios are formed every month and are held for six months. Portfolio ‘Volume’ is trading volume, ‘PESPR’ is the proportional effective spread, ‘PQSPR’ is the proportional quoted spread, and ‘Depth’ is the average depth. Depth and volume are in thousands of shares and proportional spreads are in percent. For each of these characteristics, we calculate the difference between average of these variables around the earnings announcement dates (from two days before to one day after the announcement date) and the average of these variables in a benchmark period (two weeks to one week before and one week to two weeks after the announcement date). These differences are calculated for all earnings announcement dates for upto six months after portfolio formation. The sample period is 1988 to 2004 and the portfolios include all NYSE stocks with price greater than five dollars.

Illiquidity Rank	$\Delta PQSPR$	$\Delta PESPR$	$\Delta Depth$	$\Delta Volume$
1	0.008	0.005	0.069	368.738
2	0.011	0.007	0.100	152.522
3	0.014	0.008	-0.027	100.075
4	0.014	0.010	-0.007	63.641
5	0.014	0.008	0.007	50.029
6	0.015	0.007	-0.069	22.817
7	0.016	0.014	-0.046	16.197
8	0.023	0.012	-0.162	8.257
9	0.038	0.030	-0.063	3.980
10	-0.034	0.024	0.019	2.305

Figure 1: Returns of SUE- and Illiquidity-Sorted Portfolios

This figure shows the five-year rolling averages of returns and costs of 10–1 SUE portfolio with illiquidity rank of 10. SUE is measured as $\frac{\text{absolute returns}}{\text{dollar volume}} \times 10^6$. Illiquidity is measured by the Amihud measure (ratio of absolute returns to dollar volume multiplied by 1×10^6). The breakpoints for sorts are determined by the last three months SUE and the last month Amihud measure. SUE (illiquidity) portfolio 1 has the lowest SUE (illiquidity), and portfolio 10 has the highest SUE (illiquidity). The stocks are sorted independently based on SUE and illiquidity. The portfolios are formed every month and are held for six months - the time series of portfolio returns is based on equal-weighted average of six portfolios each month. Net returns are calculated by subtracting Keim and Madhavan (1997), Korajczyk and Sadka (2004), and Chen, Stanzl, and Watanabe (2004) cost estimates. These are indicated by the mnemonics ‘KM’, ‘KS’, and ‘CSW’ respectively. The dollar amount of trading for KS and CSW estimates is five million dollars. The sample period is 1972 to 2004 and the portfolios include all NYSE and AMEX stocks with price greater than five dollars.

